EVID: Reference-free Evaluation of Action Fidelity in Videos from Robot World Models via Inverse Dynamics

Duc Anh Hoang

Abstract

Action-conditioned video world models predict the video a robot would record when it executes a given sequence of actions, and they are compared mostly with frame-level appearance metrics such as PSNR, SSIM, and LPIPS against the recorded video. These metrics leave two gaps: they need the real video, which does not exist for a new action sequence, and they score how the frames look, not whether the robot in the video performs the commanded motion or whether a real arm could perform it. We reproduce IRASim and its baselines and add Cosmos-Predict2.5 with and without action input on 105 Bridge and 57 Language-Table clips. In a controlled test where videos are generated from deliberately corrupted actions, SSIM rates a static copy of the first frame above every corrupted video and even above the video from the correct actions, and PSNR and LPIPS rate it above the three strongest corruptions, whereas an end-effector track compared with the real video degrades steadily as the corruption grows. We propose EVID, which closes the loop from actions to video and back to actions: the Cosmos3-Nano inverse dynamics model reads the end-effector trajectory from the generated video, and we measure how far it deviates from the commanded trajectory and check, with the inverse kinematics of the WidowX 250 arm, whether a real robot could execute it. Without any real video, EVID separates real videos from a static one by a factor of six, grows monotonically with the action corruption, and ranks the models in the same order as the reference-based metrics. It shows that Cosmos-Predict2.5 conditioned on actions follows them most faithfully, that the same backbone without actions moves the robot no better than a static video, and that almost every arm motion in the generated videos is executable by the real arm; the most frequent errors are instead objects that change shape or vanish, which neither EVID nor the appearance metrics detect. Code, videos, and an interactive results viewer are available at hoangducanh1865.github.io/PI.

1. Introduction

A video world model for robot manipulation takes the first camera frame I_0 and a sequence of actions $a_{1:T}$ and predicts the video $\hat{V}$ the robot would record. If the prediction is right, the model can replace a physics simulator for policy evaluation, planning, and data generation [10, 18, 20]. Such models are compared mostly with frame-level appearance metrics against the recorded video, namely PSNR, SSIM [17], and LPIPS [19], or with FVD [15] over a set of videos. These metrics leave two gaps. First, they need the real video of the same actions, which does not exist when the model generates the outcome of a new action sequence, the very use of a simulator. Second, they score how the frames look, not whether the robot in the video performs the commanded motion, whether a real arm could perform it, or whether objects move only when touched.

We reproduce IRASim [20] on Bridge [16] and generate the same 24 clips from deliberately corrupted actions: SSIM rates a static copy of the first frame above every corrupted video and even above the video from the correct actions, and PSNR and LPIPS rate it above the three strongest corruptions, whereas the end-effector track in the image, compared with the track in the real video, degrades steadily as the corruption grows. Comparing tracks, however, needs the real video, which a simulator does not have for a new action sequence. What it does have is the commanded actions, so a generated video can be checked against them instead of against a real video, which is the idea of our metric. What it does have is the commanded actions, so a generated video can be checked against them instead of against a real video, which is the idea of our metric.": What is the purpose of this claim? How does this relate to the above claims?

Our main contributions are: (1) an evaluation framework that lists, for each metric group, what it measures and what it cannot measure, with a perturbation test that checks whether a metric notices wrong actions, and (2) EVID, a reference-free metric that closes the loop from actions to video and back: an off-the-shelf inverse dynamics model reads the end-effector trajectory from the generated video, its output is converted to end-effector poses with the same formulas its authors use to prepare their training actions, so nothing is fitted, and EVID measures the deviation from the commanded

trajectory and checks whether the WidowX 250 arm could reach every pose within its joint and speed limits.

2. Related work

UniSim [18] learns a video simulator of real-world interaction from mixed internet, simulation, and robot data, generating the visual outcome of language instructions or low-level controls, and trains policies inside it that transfer to real robots. IRASim [20] generates robot videos from a first frame and a trajectory of end-effector actions with a diffusion transformer [14] that conditions each frame on its own action. It is also used to evaluate a policy, by generating videos of the policy's rollouts and letting humans label each one as a success or a failure, and to plan, by generating the outcome of several trajectories sampled from a policy and executing the one a reward model scores highest. Cosmos [10] provides pretrained video world foundation models for physical AI, and its Cosmos-Predict2.5 release includes a model post-trained on Bridge for action-conditioned generation. None of these checks, without a real video, whether the robot in a generated video performs the commanded actions.

Inverse dynamics models, which predict the action taken between two consecutive frames, are the usual way to turn videos into actions. VPT [1] trains such a model on a small labelled set to label large unlabelled video collections, UniPi [3] applies it to text-conditioned video plans to obtain robot actions, and DreamGen [5] uses it to label videos generated by a world model as training data for visuomotor policies. These works trust the actions read from the videos; we compare them with the commanded ones to evaluate the video.

VideoPhy [2] rates whether text-to-video generations follow physical commonsense with human judges and an automatic judge. Physics-IQ [9] gives a video model the first seconds of a real physical experiment, such as falling dominoes, compares where and how much motion appears in its continuation with the recorded one, and finds that visual realism does not imply physical understanding. Like our reference-based metrics, it needs the real video, and neither benchmark involves robot actions. SIMPLER [7] evaluates real-robot policies in physics simulators matched to real setups, including the WidowX setup of Bridge; it is the kind of hand-built simulator that video world models aim to replace, and a place where motions read from generated videos could be replayed with physics. Table 6 compares these sources by inputs, outputs, evaluation methods, assumptions, and limitations.

Open questions. The sources leave three questions open: whether a frame that looks real implies that the robot followed the commanded actions, whether objects obey contact by moving only when touched and keeping their identity, and whether the robot motion in the video can be executed by the real arm. We measure the first and the third with EVID, and the second only through vision-language judges.

3. Reproducing IRASim

Pipeline. IRASim [20] predicts 16 frames from the first frame and 15 actions. Each action is the end-effector motion between two frames, expressed in the end-effector frame of the earlier one: the position change $R_{t-1}^{\top}(p_t - p_{t-1})$, the Euler angles of the rotation change $R_{t-1}^{\top}R_t$, and the gripper state, with the arm part scaled by 20. The SDXL VAE encodes each 256×320 frame into a $4\times32\times40$ latent, and a diffusion transformer with patch size 2, IRASim-XL/2 [14], denoises the latents of frames 1 to 15 while the clean latent of frame 0 serves as the condition. Actions enter through adaptive layer norm with zero-initialised gates: Frame-Ada modulates the tokens of each frame by that frame's action, while Video-Ada modulates every frame by one embedding of the whole sequence. Training minimises the mean squared error between the added and the predicted noise on frames 1 to 15, with a linear schedule of 1,000 steps, 300k iterations, and learning rate 10^{-4}, and sampling uses PNDM with 50 steps. The baselines of [20] are LVDM, a latent video diffusion model with a UNet, and VDM, a diffusion model in pixel space, both conditioned on the whole action sequence.

Setup. We use the code, checkpoints, and evaluation settings of [20] at commit `c72b6da`, changing only the episode list, the seed, and the batch size. Clips are 16 consecutive frames at 5 Hz, about 3 s, from Bridge [16] test episodes, taken as the first episodes of each instruction verb: 24 core clips over 8 verbs, extended to 105 clips over 12 verbs. On Language-Table [8] we use 57 clips of 288×512 frames from 48 episodes. IRASim models run on Kaggle T4 GPUs, and the Cosmos models and the inverse dynamics model on one A100 GPU.

Reproduction details. We score a subset of the test episodes rather than the full test set, and we do not compute FVD [15], which needs far more videos than we generate. In the IRASim code the initial noise depends on the batch a clip is generated in: rerunning with the same batches reproduces the videos exactly, but a different batch composition left only 1 of 24 clips unchanged, so we treat seeds as samples and report results per seed. Our PSNR is 0.9 to 1.5 dB below the reported values on Bridge and 0.9 to 1.8 dB below on Language-Table, with the same ranking of models (Tabs. 3 and 4); the clip subset may explain the offset, which we have not verified.

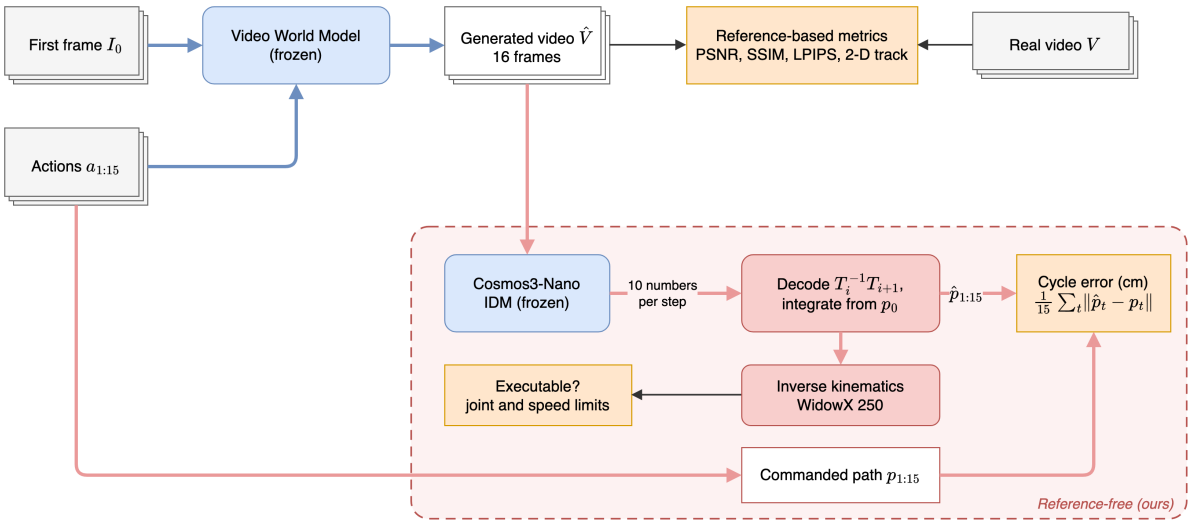


Figure 1. Evaluation pipeline. Both networks are frozen; reference-based metrics need the real video, while EVID (dashed box) needs only the commanded actions: it measures the deviation of the trajectory read back from the video and checks whether the WidowX 250 arm could execute it.

4. Evaluation framework

Models and references. Table 1 lists the inputs and outputs of the models we compare, and Fig. 1 shows how each video is scored. Cosmos-Predict2.5 [10] enters in two versions of the same 2B backbone: the model post-trained on Bridge with actions, and the base model given the first frame and the instruction text but no actions, which tests whether the action input matters. Two references bound every score: the real video of the same actions is the ceiling, and the first frame held still for 16 frames is the floor, with perfect image quality and no motion. Cosmos videos are resampled to the IRASim format of 16 frames at 256×320; for the base model we keep every third of its 49 frames at 16 fps, which spans the same 3 s.

Metric groups. Table 2 lists what each group measures and what it cannot measure. Every metric is computed per clip, and models are compared with paired Wilcoxon signed-rank tests over the same clips. The 2-D track score follows the end-effector in the image. CoTracker3 [6] tracks a 20×20 grid of points from frame 0 through the real and the generated video, and the end-effector point is the moving point whose real track is best explained by the recorded end-effector position through an affine camera; two vision-language models confirm that the point lies on the gripper. NDTW [4] compares the generated and the real track of that point,

$$\mathrm{NDTW} = \exp\left(-\frac{\mathrm{DTW}(\hat{x}_{0:15}, x_{0:15})}{16\, d_{\mathrm{th}}}\right), \qquad (1)$$

where DTW sums the pixel distances between the two tracks after aligning them in time, and $d_{\mathrm{th}} = 20$ px; NDTW is 1 for identical paths and tends to 0 for unrelated ones.

5. EVID: reading the actions back

Idea. Every metric in Tab. 2 except the judges needs the real video of the same actions, and none checks whether the motion shown could be performed by the robot. EVID uses only what a simulator always has, the commanded actions: an inverse dynamics model reads the end-effector trajectory back from the generated video, and we measure how far it deviates from the commanded trajectory and whether the arm's inverse kinematics can execute it (Fig. 1). A video that shows the commanded motion closes the cycle from actions to video and back with a small error. We use NVIDIA Cosmos3-Nano [11] in its inverse dynamics mode for Bridge, the domain `bridge_orig_lerobot`, without any training of ours.

Decoding the model output. For each pair of consecutive frames the model returns 10 numbers. Following the encoder of [12] for this domain, they are the relative pose $T_i^{-1}T_{i+1}$ of the end-effector, given as the translation and the first two columns of the rotation matrix, followed by the commanded gripper state, each quantile-normalised as

$$\hat{a} = 2\,\frac{a - q_{01}}{q_{99} - q_{01}} - 1, \qquad (2)$$

where q_{01} and q_{99} are the published 1st and 99th percentiles of each dimension. We invert Eq. (2), complete the rotation matrix with the cross product of its two columns, project it

Model	Conditioning	Output	Sampling steps
IRASim-Frame-Ada [20]	frame 0, 15 actions, one per frame	256×320, 16 frames	50
IRASim-Video-Ada [20]	frame 0, 15 actions, one embedding per video	256×320, 16 frames	50
LVDM [20]	frame 0, 15 actions, one embedding per video	256×320, 16 frames	50
VDM [20]	frame 0, 15 actions, one embedding per video	256×320, 16 frames	50
Cosmos-Predict2.5, actions [10]	frame 0, chunks of 12 actions	256×320, 13 frames per chunk	35
Cosmos-Predict2.5, base [10]	frame 0, instruction text	480×640, 49 frames at 16 fps	35

Table 1. Models compared on Bridge. The Cosmos action model generates two chunks with guidance 0, keeping frames 0 to 15; the base model uses guidance 7 and runs on the 24 core clips only, at 3 min per clip on an A100.

Group and metrics	Measures	Cannot measure
Visual quality: PSNR, SSIM [17], LPIPS [19] against the real clip	how close each frame looks to the recorded frame	whether the robot moved as commanded, since a still video keeps a high SSIM and small shifts are punished like large errors
Temporal consistency: DINOv2 [13] similarity of adjacent frames and to frame 0, optical-flow warp error	smoothness and identity stability across frames	correct motion, since the still video scores perfectly
2-D track: CoTracker3 [6] end-effector track against the real clip, NDTW [4], pixel error	whether the gripper moves along the same image path as in the real clip	depth and orientation; it needs the real clip and an end-effector point checked by vision-language models
Physical plausibility and outcome: two vision-language judges	violations a judge sees, such as objects that vanish or change and a robot that deforms, and whether the final state matches the real clip	a reliable verdict, since the judges differ in strictness and change answers between runs
Action cycle consistency: EVID, ours	whether the 3-D end-effector trajectory read from the video matches the commanded one, in cm, and whether the arm could execute it, without a real video	object motion and contact, collisions and torques; it is bounded by the error of the inverse dynamics model, 1.50 cm on real clips

Table 2. Metric groups. Task completion is covered by the outcome question of the judges.

onto the nearest rotation, and chain the relative poses from the true first pose T_0, expressed in the flange frame of the encoder, which gives the trajectory $\hat{p}_{1:15}$ read from the video; nothing is fitted. The Bridge states used by IRASim differ from the LeRobot version used by NVIDIA by a roll of π, with a median roll of 2.47 rad in our clips against 0.00 in LeRobot, so we set $R_{\text{LeRobot}} = R_{\text{IRASim}}\, R_x(\pi)$. Encoding our states with the code of [12] and decoding them with ours returns them within $2 \cdot 10^{-4}$ mm.

Cycle error. The cycle error of a clip is the mean distance between the end-effector positions read from the video and the commanded ones,

$$E_{\text{cycle}} = \frac{1}{15} \sum_{t=1}^{15} \|\hat{p}_t - p_t\|_2, \tag{3}$$

in centimetres, lower being better.

Executability. We solve the inverse kinematics of the WidowX 250 6-DOF arm, the Bridge robot, for each of the 16 poses read from the video, following the Interbotix driver: the screw axes and home pose of the arm, Newton iterations with tolerances of 1 mrad and 1 mm from three default initial guesses, the joint limits, and the joint velocity limit of π rad/s, checked between consecutive poses 0.2 s apart at the 5 Hz Bridge control rate. Newton's method from the three default guesses misses valid solutions for low, steeply pitched poses, rejecting 4 of the 105 recorded Bridge trajectories, so we add a grid of 125 initial guesses with the same tolerances and limits. A clip is executable when every pose has a solution within the joint limits and every step stays below the velocity limit. With the wrong roll convention, none of the 384 recorded poses of the 24 core clips has a solution, so the check is sensitive to the frame convention.

Use for selection. Because EVID needs no real video, it can rank several samples generated from the same actions and keep the one with the lowest cycle error, which we test as best-of-N selection.

6. Results

Bridge, 105 clips. Among the IRASim models, Frame-Ada leads, Video-Ada is second, and LVDM third (Tab. 3). Frame-Ada beats Video-Ada on PSNR with $p = 2 \cdot 10^{-5}$, better on 69% of clips, and on 2-D NDTW with $p = 0.02$, and it beats LVDM on PSNR, LPIPS, and NDTW with $p < 10^{-8}$; its two seeds do not differ, with p from 0.15

Model	Visual			2-D track		3-D	Reported
	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	px↓	cm↓	PSNR↑
IRASim [20] and its baselines							
Frame-Ada, seed 3407	23.95	0.809	0.082	0.699	10.6	1.81	**25.28**
Frame-Ada, seed 3408	23.78	0.807	0.084	0.719	9.2	1.84	
Video-Ada	23.52	0.804	0.087	0.682	10.5	1.95	24.73
LVDM	22.62	0.788	0.103	0.593	14.9	2.54	23.55
Cosmos-Predict2.5 [10]							
With actions	**25.74**	**0.859**	**0.063**	**0.754**	**7.4**	**1.64**	-
References							
Real video	-	-	-	1	0	1.50	-
Still first frame	19.80	0.805	0.141	0.156	58.7	9.06	-

Table 3. Bridge, 105 clips. Visual metrics over frames 1 to 15; 2-D track on the 96 clips with an end-effector point; 3-D: cycle error of EVID. Reported: PSNR in [20] on its test set. Best in bold, second underlined, references excluded.

Model	Visual			2-D track		Reported
	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	px↓	PSNR↑
IRASim [20] and its baselines						
Frame-Ada, seed 3407	27.03	0.861	**0.059**	**0.715**	**7.9**	**28.82**
Frame-Ada, seed 3408	**27.27**	0.862	**0.059**	0.688	10.6	
Video-Ada	23.04	0.833	0.104	0.304	37.2	23.89
LVDM	26.44	**0.863**	0.066	0.646	10.9	28.25
Reference						
Still first frame	22.35	0.879	0.103	0.131	80.7	-

Table 4. Language-Table, 57 clips. Columns as in Tab. 3; best in bold, second underlined, reference excluded.

to 0.4. Cosmos-Predict2.5 with actions is the best model on every metric: against Frame-Ada it gains 1.78 dB PSNR with $p = 4 \cdot 10^{-16}$ and 0.055 NDTW with $p = 0.008$, and lowers the cycle error by 0.16 cm with $p = 0.02$, to within 0.14 cm of the error on the real videos. SSIM separates no model from the still video.

Language-Table, 57 clips. On table-top pushing in 2-D (Tab. 4), Frame-Ada again beats LVDM on PSNR with $p = 0.015$ and on NDTW with $p = 0.017$, and SSIM ranks the still frame first, 0.879 against 0.861 to 0.863, because most of the image is a static table. Video-Ada is far behind on this dataset, below LVDM on every metric with $p < 0.001$ and close to the still floor on motion; [20] reports the same gap, 23.89 dB for Video-Ada against 28.25 for LVDM and 28.82 for Frame-Ada, so it is a property of the model rather than of our run. EVID is not available here, because the inverse dynamics model has no Language-Table domain, and Cosmos-Predict2.5 provides no action-conditioned model for Language-Table.

Perturbation test. To check whether a metric notices wrong actions, we regenerate the 24 core clips with Frame-Ada from corrupted actions, namely random walks of 0.5 and 1.5 cm per frame on the position, mirrored motion, and all actions set to zero, and score them against the real clips of the original actions (Tab. 5). NDTW and the cycle error of EVID both degrade steadily as the corruption grows, place zero actions at the level of the static video, 0.133 against 0.135 and 9.73 against 9.52 cm, and place mirrored motion beyond it; EVID does so without the real video. The appearance metrics do not: SSIM rates the static video above every corrupted generation and even above the generation from the original actions, and PSNR and LPIPS rate it above the three strongest corruptions, so a higher appearance score does not mean that the robot followed the actions.

Actions given	Visual			2-D	EVID	
	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	cm↓	Exec.↑
IRASim-Frame-Ada, seed 3407						
Original	**23.97**	**0.814**	**0.081**	**0.759**	**1.93**	**100**
Noise 0.5 cm	21.87	0.788	0.103	0.550	2.67	**100**
Noise 1.5 cm	19.45	0.752	0.149	0.270	5.62	**100**
All zero	19.31	0.753	0.158	0.133	9.73	**100**
Mirrored	18.59	0.737	0.178	0.036	15.32	83
Reference						
Still first frame	20.35	0.816	0.134	0.135	9.52	100

Table 5. Perturbation test on the 24 core clips, scored against the real clip of the original actions; NDTW on the 13 clips whose end-effector point both judges confirmed; Exec.: percentage of clips whose motion read from the video is executable by the WidowX 250. Best in bold, second underlined, reference excluded.

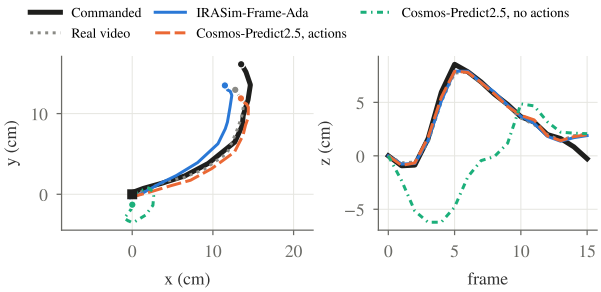


Figure 2. EVID on the clip "take banana out of colander": end-effector path read back from each video against the commanded path, from above (left) and as height over time (right). Cycle errors: real video 1.63 cm, Frame-Ada 1.86, Cosmos-Predict2.5 with actions 1.82, without actions 15.53.

Validating the cycle error. On the 105 real clips, the relative motion read by the inverse dynamics model correlates with the recorded one, with r of 0.92, 0.93, and 0.88 for x, y, and z and 0.81 to 0.92 for the rotation entries, and the cycle error is 1.50 cm, with a 95% bootstrap interval of 1.36 to 1.68, for clips that move 25 cm on average. On the still video it is 9.06 cm, as bad as assuming no motion

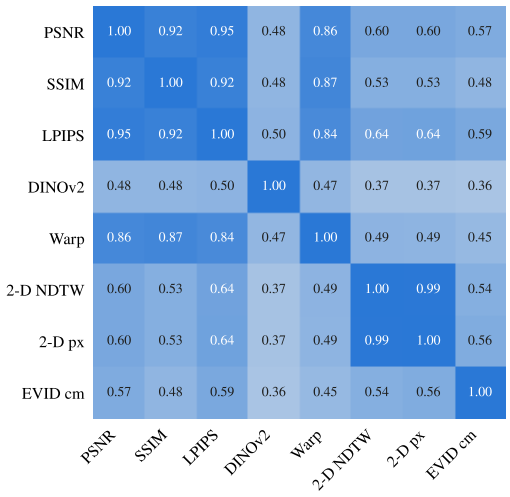


Figure 3. Spearman correlation between metrics over the 502 generated Bridge clips scored by every metric, each oriented so that higher is better.

at 9.32 cm, so the model reads the motion from the frames rather than from a prior. The gripper command is read with a balanced accuracy of 0.87, against 0.50 for always predicting an open gripper. Frame-Ada scores 1.81 and 1.84 cm over two seeds, worse than the real videos with $p < 10^{-5}$, yet better on 27 to 29% of clips, because the error of the inverse dynamics model on a clip is shared by the real and the generated video, with a Spearman correlation of 0.81 between the two. Its per-clip score correlates with LPIPS at $\rho = 0.52$, but the part exceeding the score of the real video barely does, at $\rho = 0.21$ for LPIPS and 0.09 for NDTW, so the cycle error measures something the pixel and image-track scores do not. On the 24 core clips it ranks the models in the same order as the reference-based scores: Frame-Ada 1.86 to 2.12 cm over six seeds, Video-Ada 2.09 and 2.13 over two seeds, LVDM 2.15 and 2.53, and VDM 10.68. Figure 2 shows one clip: the paths read from the real video and from both action-conditioned models stay within about 2 cm of the commanded one, while the model without actions moves the arm elsewhere.

Metric agreement. Over the 502 generated clips scored by every metric (Fig. 3), the appearance metrics PSNR, SSIM, LPIPS, and the warp error form one tight group, with Spearman correlations of 0.84 to 0.95, which the trajectory metrics join only moderately, at 0.45 to 0.64. The cycle error of EVID, computed without the real video, correlates as much with the 2-D track, 0.54 and 0.56, as with the appearance metrics, 0.48 to 0.59. All correlations are positive partly because a hard clip is hard for every metric; the perturbation test (Tab. 5) shows where the two groups disagree.

Executability. The check passes 104 of the 105 recorded Bridge trajectories; the one rejected brings the forearm roll joint to its limit of $\pm\pi$, so about 1% of clips are false alarms. On the motion read from the videos, 99.0% of clips are executable for the real videos and for every action-conditioned model, the one failure being the same clip in all of them, where the inverse dynamics model reads poses out of the arm's reach even from the real video; Cosmos-Predict2.5 without actions passes 23 of 24 clips, and the static video passes all of them. In the perturbation test only mirrored motion fails, on 17% of clips (Tab. 5). The arm motion shown by these models is therefore almost always executable, and the check separates only grossly wrong motion: executability is a necessary condition, while the cycle error is what separates the models.

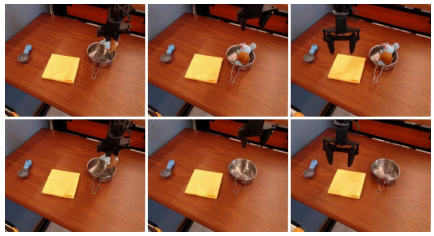
Action input matters. On the 24 core clips, Cosmos-Predict2.5 base, given the first frame and the instruction but no actions, produces smooth videos, with a DINOv2 adjacent-frame similarity of 0.916 against 0.953 for real clips, whose robot does not follow the recorded motion: its cycle error of 10.30 cm does not differ from the still video at 9.52 cm ($p = 0.32$), against 1.74 cm for the same backbone with actions ($p = 10^{-7}$). Its PSNR of 13.5 dB is also far below every action-conditioned model, partly because of resampling its 49 frames at 16 fps to our format. On the clip "move the toy and put it in the vessel" it even replaces the WidowX with a different, white robot arm, a failure of embodiment that the commanded actions prevent.

Temporal consistency and judges. On the 24 core clips, DINOv2 adjacent-frame similarity is 0.953 for real clips and 0.924 to 0.938 for the latent models, while the still video scores 1.000, so these metrics measure stability rather than correct motion. Two vision-language judges, gemini-3.8-flash and gpt-6.1-sol, rate physical plausibility and whether the final state matches the real clip; they agree on the ranking but not on strictness, flagging 21% and 4% of real clips. The most frequent violation for every latent model is an object that changes or vanishes, in 46% and 17% of Frame-Ada clips, and the still video is judged plausible every time, caught only by the outcome question with 0.00 and 0.04.

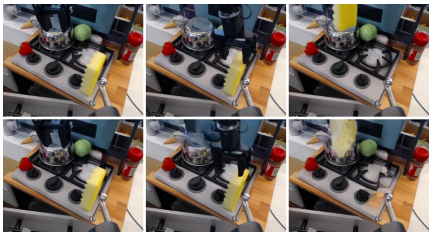
Best-of-N selection. For each of the 24 core clips we generate six Frame-Ada videos and keep the one with the lowest cycle error. On the independent reference-based scores the selection does not help measurably: PSNR is 24.02 against 23.93 dB for the average seed ($p = 0.64$) and NDTW 0.717 against 0.709 ($p = 0.37$), while an oracle choosing by PSNR itself, which needs the real video, would reach 24.86 dB. Seeds of Frame-Ada differ little, so there is little to select from; the selection may matter for models whose samples differ more.



(a) "put eggplant in pot": success
PSNR 23.33, cycle error 1.18 vs 1.00 cm



(b) "put the toy in the vessel": the toy vanishes
PSNR 24.55, cycle error 2.57 vs 2.52 cm



(c) "wipe pot with sponge": the gripper dissolves
PSNR 22.27, cycle error 2.95 vs 1.80 cm

Figure 4. Frames 0, 8, and 15 of the real video (top rows) and of IRASim-Frame-Ada from the same actions (bottom rows); both judges agree on each verdict. The cycle errors of EVID are given for the generated and the real video. In (b) the arm follows the actions and the PSNR is the highest of the three, yet the object vanishes, which neither metric detects; in (c) the contact failure also shows in the cycle error.

7. Strengths and limitations of IRASim

Strengths. *S1, the robot follows the actions.* The gripper of Frame-Ada follows the real image path, with NDTW 0.77 and a mean error of 6.3 px on verified clips, and the 3-D path read back from its videos is within 1.81 cm of the commanded one, against 1.50 cm for the real videos. *S2, per-frame conditioning helps.* With the same backbone and data, modulating each frame by its own action in Frame-Ada beats one embedding of the whole sequence in Video-Ada on PSNR and 2-D tracking over 105 paired clips. *S3, stable and reproducible.* Temporal consistency is close to the real videos, two seeds give statistically equal scores, and a rerun with the same batches reproduces the videos exactly.

Limitations. *L1, objects do not keep their identity.* The most frequent violation found by the judges is an object that changes shape or vanishes, typically once the gripper holds it (Fig. 4b). *L2, correct motion does not imply the correct outcome.* Only 50% and 75% of Frame-Ada clips end in the same state as the real clip according to the two judges. *L3, errors grow with contact and time.* The gap between the cycle error of Frame-Ada and that of the real video is largest for wipe (Fig. 4c), turn, slide, and topple, at 0.5 to 0.7 cm, and close to zero for open, flip, and fold; within a clip the error grows towards the last frames. *L4, short clips of one embodiment.* Each checkpoint covers one dataset at 256×320 and 16 frames, about 3 s; longer horizons need autoregressive chaining, which we do not test.

8. Conclusion

On real robot data, frame-level appearance metrics cannot tell whether the robot in a generated video followed its actions: SSIM prefers a static video even to the generation from the correct actions, while trajectory-based scores degrade as the actions get worse. EVID provides this check without any real video, by reading the actions back with an off-the-shelf inverse dynamics model decoded exactly as its authors encode them: IRASim-Frame-Ada follows Bridge actions to within about 0.3 cm of what the same model reads from real videos, Cosmos-Predict2.5 with actions comes closer still, and the same backbone without actions moves the robot no better than a static frame. Almost every arm motion read from the generated videos is executable by the WidowX 250, so these models fail on objects, which change or vanish, rather than on the arm. Selecting the best of six IRASim samples by the cycle error did not improve independent scores, since the samples differ little. The study is limited to a few hundred clips, one embodiment for EVID, judges that are not stable between runs, and an inverse dynamics model whose own error of 1.50 cm bounds what EVID can resolve and whose training data may include Bridge test episodes.

References

[1] Bowen Baker, Ilge Akkaya, Peter Zhokhov, Joost Huizinga, Jie Tang, Adrien Ecoffet, et al. Video PreTraining (VPT): Learning to act by watching unlabeled online videos. *arXiv preprint arXiv:2206.11795*, 2022.

[2] Hritik Bansal, Zongyu Lin, Tianyi Xie, Zeshun Zong, Michal Yarom, Yonatan Bitton, et al. VideoPhy: Evaluating physical commonsense for video generation. *arXiv preprint arXiv:2406.03520*, 2024.

[3] Yilun Du, Mengjiao Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. *arXiv preprint arXiv:2302.00111*, 2023.

[4] Gabriel Ilharco, Vihan Jain, Alexander Ku, Eugene Ie, and Jason Baldridge. General evaluation for instruction conditioned navigation using dynamic time warping. *arXiv preprint arXiv:1907.05446*, 2019.

[5] Joel Jang, Seonghyeon Ye, Zongyu Lin, Jiannan Xiang, Johan Bjorck, Yu Fang, et al. DreamGen: Unlocking generalization in robot learning through video world models. *arXiv preprint arXiv:2505.12705*, 2025.

[6] Nikita Karaev, Iurii Makarov, Jianyuan Wang, Natalia Neverova, Andrea Vedaldi, and Christian Rupprecht. CoTracker3: Simpler and better point tracking by pseudo-labelling real videos. *arXiv preprint arXiv:2410.11831*, 2024.

[7] Xuanlin Li, Kyle Hsu, Jiayuan Gu, Karl Pertsch, Oier Mees, Homer Rich Walke, et al. Evaluating real-world robot manipulation policies in simulation. *arXiv preprint arXiv:2405.05941*, 2024.

[8] Corey Lynch, Ayzaan Wahid, Jonathan Tompson, Tianli Ding, James Betker, Robert Baruch, Travis Armstrong, and Pete Florence. Interactive language: Talking to robots in real time. *arXiv preprint arXiv:2210.06407*, 2022.

[9] Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? *arXiv preprint arXiv:2501.09038*, 2025.

Source	Inputs	Outputs	Evaluation	Assumptions	Limitations
Video world models					
UniSim [18]	observation, language instruction or low-level control	video of the visual outcome	FID, FVD, IS, CLIP score in ablations; policies trained purely in the simulator and deployed zero-shot on real robots	mixing internet, simulation, and robot data yields one simulator of real interaction	hallucinated outcomes for unrealistic actions and limited memory, as stated by the authors
IRASim [20]	first frame, end-effector action trajectory	16 frames at 256×320	PSNR, SSIM, latent L2, FID, FVD against recorded videos; human preference; policy evaluation on LIBERO with human-judged success, Pearson 0.99 with the simulator; model-based planning	closeness to the recorded video measures simulation quality	action following is checked by frame metrics or by humans judging task success; one embodiment per checkpoint
Cosmos [10]	video and text; actions after post-training	video	3-D consistency by epipolar error and pose estimation; physics alignment by PSNR, SSIM, DreamSim, IoU on simulated scenes	pretraining on large video data transfers to physical AI after post-training	physics alignment measured on simulated scenes
Video models turned into actions by inverse dynamics					
UniPi [3]	first frame, text goal	video plan, then actions from an inverse dynamics model	task success in simulated manipulation; on Bridge, CLIP score, FID, FVD, and a success classifier on the last generated frame	a generated plan is executable and its actions can be recovered	slow diffusion sampling, as stated by the authors; plan errors reach the robot unchecked
DreamGen [5]	first frame, language, target robot	video, then pseudo-actions from a latent action or inverse dynamics model	DreamGen Bench with instruction following and physics alignment by VLM and human judges; downstream policy success	pseudo-actions from generated videos are useful training data	the benchmark scores the video, not the recovered actions
Evaluation of physical plausibility and of policies					
VideoPhy [2]	text prompt	text-to-video generation	human judgement of semantic adherence and physical commonsense; automatic judge VideoCon-Physics	physical commonsense can be judged from the video alone	no actions or robots; best model correct on 39.6% of prompts
Physics-IQ [9]	frames of a real physical experiment	predicted continuation	spatial and spatiotemporal IoU, weighted spatial IoU, and MSE of motion against the recorded continuation	the real continuation is the reference	needs the real video; no robot actions
SIMPLER [7]	robot policy	rollouts in simulated scenes matched to real setups	correlation of simulated and real success rates	reducing control and visual gaps suffices without full digital twins	evaluates policies rather than videos; few real setups

Table 6. Literature matrix. Evaluation methods are taken from each paper.

[10] NVIDIA. Cosmos world foundation model platform for physical AI. *arXiv preprint arXiv:2501.03575*, 2025.

[11] NVIDIA. Cosmos3-Nano model card. https://huggingface.co/nvidia/Cosmos3-Nano, 2026. Accessed 2026-10-29.

[12] NVIDIA. Cosmos framework. https://github.com/NVIDIA/cosmos-framework, 2026. Commit 1a2f03a, accessed 2026-10-29.

[13] Maxime Oquab et al. DINOv2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[14] William Peebles and Saining Xie. Scalable diffusion models with transformers. *arXiv preprint arXiv:2212.09748*, 2022.

[15] Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphael Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.

[16] Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, Andre He, Vivek Myers, Kuan Fang, Chelsea Finn, and Sergey Levine. BridgeData V2: A dataset for robot learning at scale. *arXiv preprint arXiv:2308.12952*, 2023.

[17] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[18] Sherry Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive real-world simulators. In *International Conference on Learning Representations*, 2024.

[19] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *CVPR*, 2018.

[20] Fangqi Zhu, Hongtao Wu, Song Guo, Yuxiao Liu, Chilam Cheang, and Tao Kong. IRASim: A fine-grained world model for robot manipulation. *arXiv preprint arXiv:2406.14540*, 2024.